

WILDENRATH, Germany

WMO: 104020

Lat: 51.117N

Lon: 6.217E

Elev: 89

StdP: 100.26

Time Zone: 1.00 (EUC)

Period: 82-91

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.2	-7.5	-14.1	1.1	-9.3	-11.2	1.5	-6.1	13.2	8.8	11.9	7.6	2.8	60	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	28.9	19.5	27.2	18.7	25.3	18.0	20.5	26.9	19.6	25.5	18.7	23.9	3.2	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.4	22.8	17.5	12.6	21.8	16.7	12.0	21.1	59.4	27.2	56.3	25.7	53.2	24.0	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	9.0	7.9	DB	-11.4	32.6	5.4	2.0	-15.3	34.0	-18.5	35.2	-21.5	36.3	-25.5	37.8	(4)
(5)			WB	-12.0	22.4	5.2	0.8	-15.7	23.0	-18.8	23.4	-21.7	23.9	-25.5	24.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	2.1	1.8	5.8	8.5	12.8	15.3	18.1	17.8	14.6	11.3	5.9	4.1	(6)	
	DBStd	6.76	5.07	4.73	3.51	3.51	3.64	3.54	3.27	2.92	3.00	3.24	3.70	3.84	(7)	
	HDD10.0	1034	244	229	138	69	14	1	0	0	1	22	128	187	(8)	
	HDD18.3	3172	502	462	390	294	174	106	47	47	119	218	372	442	(9)	
	CDD10.0	997	1	1	6	25	101	161	250	241	138	64	7	3	(10)	
	CDD18.3	92	0	0	0	0	3	14	38	29	6	1	0	0	(11)	
	CDH23.3	859	0	0	0	1	41	133	362	268	49	3	0	0	(12)	
	CDH26.7	204	0	0	0	0	1	24	102	70	7	0	0	0	(13)	
(14) Wind	WSAvg	3.3	4.4	3.8	3.8	3.1	2.7	2.9	2.8	2.7	2.9	3.5	3.6	3.8	(14)	
(15) Precipitation	PrecAvg	761	63	59	50	44	58	72	75	79	59	62	67	73	(15)	
	PrecMax	974	136	137	93	103	112	189	163	162	159	154	108	157	(16)	
	PrecMin	597	6	10	3	1	17	14	32	18	12	34	6	26	(17)	
	PrecStd	95	30	30	25	23	26	37	35	43	37	24	25	31	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	15.1	18.8	22.6	26.5	29.0	31.8	31.8	27.9	23.2	16.0	14.0	(19)	
		MCWB	9.9	10.1	11.9	14.0	16.1	19.5	20.6	19.3	18.6	17.2	12.5	10.9	(20)	
	2%	DB	10.6	12.2	15.1	19.8	24.6	27.1	29.3	28.5	24.6	20.1	13.9	12.4	(21)	
		MCWB	8.7	9.4	10.5	11.9	15.8	18.7	20.2	19.0	17.9	15.2	11.4	10.1	(22)	
	5%	DB	9.4	9.7	12.5	17.3	22.4	24.6	27.5	26.3	22.1	17.8	12.5	10.9	(23)	
		MCWB	8.0	7.8	9.3	11.2	14.5	17.5	19.3	18.5	16.7	14.2	10.7	9.2	(24)	
	10%	DB	8.4	8.1	10.8	15.1	20.1	22.1	25.2	24.1	20.0	16.2	11.3	9.4	(25)	
		MCWB	7.1	6.5	8.4	9.9	13.9	16.8	18.6	17.4	15.9	13.2	9.7	8.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	11.3	12.3	14.9	17.9	21.4	22.0	20.9	20.2	17.7	13.2	11.8	(27)	
		MCDB	11.7	14.0	17.2	21.7	24.4	26.2	29.2	28.2	25.3	22.1	14.8	12.8	(28)	
	2%	WB	9.2	9.5	11.1	13.0	16.4	19.7	20.9	19.8	18.7	15.8	12.0	10.3	(29)	
		MCDB	10.0	11.5	14.6	17.4	22.4	25.0	27.8	26.2	23.0	18.6	13.5	11.8	(30)	
	5%	WB	8.2	8.1	9.9	11.4	15.2	18.4	20.0	19.0	17.5	14.7	10.9	9.4	(31)	
		MCDB	9.1	9.6	11.7	16.2	21.0	23.1	26.1	25.1	20.8	17.2	12.2	10.7	(32)	
	10%	WB	7.1	6.6	8.9	10.4	14.3	17.0	18.9	18.1	16.4	13.7	9.9	8.4	(33)	
		MCDB	8.5	8.0	10.7	14.7	19.3	20.9	24.1	22.9	19.5	15.9	11.3	9.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	6.3	7.3	9.1	9.3	8.7	9.5	9.9	8.6	7.3	5.6	4.7	(35)	
		MCDBR	5.3	8.0	9.9	12.8	13.4	12.7	13.4	13.7	12.4	10.0	7.2	5.4	(36)	
	5% WB	MCWBR	4.7	6.1	6.4	6.7	6.6	6.1	6.0	6.0	6.5	6.1	5.3	4.5	(37)	
		MCWBR	5.2	7.6	8.7	11.7	11.8	11.7	11.9	12.2	10.9	8.9	6.5	5.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.323	0.346	0.377	0.412	0.418	0.418	0.424	0.415	0.387	0.361	0.337	0.319	(40)	
		taud	2.410	2.370	2.289	2.214	2.237	2.254	2.236	2.285	2.353	2.419	2.408	2.380	(41)	
	Ebn at Noon		696	775	814	825	838	840	826	814	796	748	668	633	(42)	
		Edh at Noon	62	82	106	128	131	130	131	118	99	78	61	55	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.75	1.42	2.56	4.10	4.91	5.32	5.04	4.31	3.18	1.87	0.88	0.57	(44)	
	RadStd		0.11	0.27	0.38	0.47	0.45	0.47	0.62	0.41	0.36	0.17	0.11	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (86 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page